

Out of Time

In-Class Assignment

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The stat system call is useful for querying attributes of a file. Specifically, we can access the modified, access, and changed time of a file. The modified time records when the file was last modified, access time records when it was last accessed, and changed time records when the file status was last changed (i.e., inode data).

Create a new file in a folder of your choosing titled 'myTimeTest.c'.

myTimeTest.c:

```
#include <stdio.h>
#include <time.h>
#include <sys/stat.h>
#include <utime.h>
```

```
const char *filename = "myTestFile.txt";
```

```
int main()
{
    return 0;
}
```

Create a variable 'myStat' which is a structure of type 'stat'.

Using the touch command, create a new file 'myTestFile.txt'. You can do this by typing 'touch myTestFile.txt'. This file will be used to examine the aforementioned times (modified, accessed, changed).

Add some lines of code to myTimeTest.c that will output each of the above times after the stat function is called. The printf statements are listed below, but you will have to determine the correct stat system call to make:

```
// insert stat system call

printf("Modified \t%s", ctime(&myStat.st_mtime));
printf("Accessed \t%s", ctime(&myStat.st_atime));
printf("Changed \t%s", ctime(&myStat.st_ctime));
```

Combining the 'utime' command and 'stat' command you are able to change the 'modified' and 'accessed' time of a file. For this particular exercise you will need to roll each of these dates back by exactly one day. Do some research to determine how the utime system call works (<https://man7.org/linux/man-pages/man2/utime.2.html>) (Note: 'utimes' isn't supported on all systems so please use the 'utime' system call). After the utime system call (and any other necessary code to make this call), you will need to make another call to stat to update the filesystem information with the new time. Print out the values of each of the variables after the time has been rolled back a day for the modified and accessed time.

Once the program is complete you should be able to type 'ls -l' and note that you actually created the file 'myTestFile.txt' yesterday and not today...

Hand-in:

A screenshot of your program execution

The final source code used to accomplish the assignment objectives (myTimeTest.c)